HOVER OS · AN ESSAY

AI Is the Catalyst for the Next Input Era — and the Next Trillion-Dollar Platform

An essay on input paradigms, platform value, and the case for Hover OS.

AT A GLANCE

Five points, if you read nothing else.

The market has always rewarded whoever owns how humans express intent to machines. AI is shifting that layer again — and the most defensible place to build for it is the shared physical space.

01 **Input paradigms dictate operating system value** — not raw compute.

02 **AI is the new input translation layer** between human intention and machine execution.

03 Big tech is stuck on single-user personal devices. **Our white space is shared, multi-user physical space.**

04 The addressable software market is **$45B–$70B**, verified by a bottom-up check of real-world spaces.

05 **We're sensor-agnostic**, using reference hardware simply to break adoption friction.

The market has historically rewarded the platform that owns how humans express intent to machines. Compute changes matter — faster chips, bigger models, cheaper inference — but they are infrastructure. The operating system layer captures disproportionate value only when a new input modality becomes dominant. We believe AI is creating exactly that shift, and that the most defensible place to build for it is not the personal device, but the shared physical space.

01 · HISTORY

The historical pattern: platforms win by owning intent

Look back across the major eras of computing and a consistent pattern emerges. Each era was not defined primarily by its processors, its networks, or even its software catalog. It was defined by how humans expressed intent to a machine, and by which company built the layer that translated that expression into action. Every time the primary interaction model changed, a new operating system emerged to own it, and that ownership, rather than the underlying hardware, is where the value concentrated.

The mainframe era came first, and its input was physical. Users expressed intent through punched cards and magnetic tape — preparing decks of instructions, handing them to an operator, and waiting hours for a batch job to return. That interaction model created the first commercial computing market, and the platform that owned it was IBM: System/360 and its operating system, OS/360, standardized how programs and data entered the machine across an entire product line. The mainframe never went away — it remains a market of roughly $5 billion a year and still runs much of the world's core banking and transaction processing — but an input model built around batches and operators kept computing at the scale of institutions rather than individuals.

The desktop era that followed was built around the keyboard and the mouse. Users expressed intent through typed commands, pointers, windows, and menus, and that interaction model created the modern personal computer market — a category now worth on the order of $235 billion annually. The winning platform in that era was not simply the best-built computer. It was Windows, the software layer that stood between human intent and machine action, and it captured a wildly disproportionate share of the value created.

The smartphone revolution repeated the pattern, and it is worth being precise about what actually drove it: it was not primarily a story about faster mobile processors. It was a new input story. Humans moved from clicking and typing to touching, swiping, pinching, and gesturing directly on glass, and the market that resulted grew to roughly $609 billion annually — about 2.6 times the size of the desktop market it grew alongside, not replaced. Once again, value concentrated at the operating system layer. iOS and Android became the control points of mobile computing not because they made the best silicon, but because they owned the interaction paradigm.

"When the input method changes, the platform changes."

THE LESSON OF SIXTY YEARS OF COMPUTING HISTORY

02 · THE SHIFT

AI as the fourth input revolution

We believe artificial intelligence is creating the fourth major input paradigm, and we think most of the market is looking at it the wrong way. The prevailing discussion frames AI as a compute workload — larger models, faster inference, bigger datacenters, more GPUs. All of that matters, but it describes infrastructure. It does not explain where platform value ultimately accumulates.

"Computing does not begin with computation. Computing begins with human intent."

For the first time, computers can understand voice, gesture, gaze, presence, movement, spatial context, and environmental awareness, all without a keyboard, a mouse, or a pane of glass in between. The interface becomes natural rather than mechanical: people stop telling a machine what to do, one keystroke or tap at a time, and start interacting with an intelligence through words and gestures, much as they would with another person. The user no longer adapts to the computer; the computer adapts to the user. That is the real significance of this moment: **AI is becoming the translation layer between human intention and machine execution**, in the same way that a graphical shell once translated keystrokes into commands, and a touch layer once translated finger movement into action.

The input surface will keep widening, and that matters for how this layer should be designed. Brain–computer interfaces are the clearest next entrant: EEG work is already modelling thought patterns with useful reliability, and the remaining barrier is sensitivity at a distance rather than inference. Today BCI requires a worn device; once detection at even a metre becomes practical, it becomes one more stream of human intent that has to be translated into machine action. A platform whose function is translating natural human signals into system commands should not have to be rebuilt to absorb a new signal — which is an argument for a sensor-agnostic architecture from the outset, and for treating input modality as a plug-in rather than a premise.

Markets expand; they rarely divide

History also offers a clear pattern about what happens to market size when the input layer changes, and it argues against the zero-sum framing that dominates most spatial computing coverage. Mainframes did not disappear when the personal computer arrived. Desktop did not disappear when mobile arrived. Mobile did not eliminate desktop. Each new input model created an entirely new category of usage layered on top of the ones that came before it, and we expect that pattern to repeat as spatial and physical AI systems come online.

COMPUTING ERA	PRIMARY INPUT	MARKET SIZE	PLATFORM WINNER
Mainframe	Punch cards + tape	~$5B	IBM (System/360)
Desktop	Keyboard + mouse	~$235B	Windows
Mobile	Multi-touch	~$609B	iOS / Android
Spatial & physical AI	Voice + gesture + presence	**~$1.2T** (2035E)	**Unclaimed**

The projected $1.2 trillion spatial computing market is not a redistribution of the desktop or mobile markets. It is the next expansion of human-computer interaction, arriving on schedule with the same shape as the three before it — a new way of expressing intent, growing into a market roughly double the one it follows.

Where the $1.2 trillion actually sits

A headline market number is only useful if it can be decomposed into slices a company can charge for. Spatial and physical AI resolves into six subsegments, and they differ enormously in size, margin structure, and whether an operating layer can capture any of

them. The allocations below are our own analytic split rather than a sourced third-party breakdown; we present them because the shape matters more than the precision.

SUBSEGMENT	SHARE OF 2035E TAM	HOVER OS POSITION
Output / display & XR devices	~$350B 29%	Not addressable — an output surface we attach to
Embodied AI / robotics	~$350B 29%	Adjacent — we read the room, robots act in it
Sensing / capture	~$150B 12%	Partially addressable via first-party devices (~$8–12B)
Interpretation / intent models	~$120B 10%	An input we license or embed, not a slice we own
Digital twin / environment modelling	~$120B 10%	Adjacent — twins describe spaces, Hover OS operates them
Platform / spatial OS	**~$110B 9%**	**Core market — ~40% of it is shared-space, unserved by incumbents**

Two thirds of the headline number is device and robotics hardware — unit-driven, capital-intensive, and already contested by Apple, Meta, and the humanoid robotics cohort. The platform layer is the smallest slice at roughly nine percent. It is also the slice that has historically captured the majority of an era's equity value: Windows was a fraction of PC hardware revenue and still became the most valuable asset of the desktop era. Within that nine percent, personal-device spatial operating systems account for the majority and are structurally closed to us. The shared and ambient portion — roughly forty percent of the segment — is where no incumbent is architected to compete, because every one of them assumes one user, one device, one session.

ECOSYSTEM WE ATTACH TO

$1.2T

Spatial & physical AI, 2035E

MARKET WE CAN PRICE INTO

$45–70B

Core shared-space platform, plus hardware attach and app revenue

BOTTOM-UP CHECK

$20–75B

4–5M shared spaces × $5–15K per space per year

That puts our core addressable market near $45 billion by 2035, rising to roughly $55–70 billion once first-party hardware attach and platform-mediated application revenue are included. A bottom-up check lands in the same range: there are on the order of four to five million commercially addressable shared spaces globally — museums and attractions, retail, healthcare facilities, classrooms, transport concourses, corporate collaboration space, industrial floors — and at five to fifteen thousand dollars per space per year in platform licensing, that is a twenty to seventy-five billion dollar annual market. We would rather defend a narrow number with a bottom-up check than a trillion-dollar headline. The first five years of deployments address low single-digit billions of it.

04 · OPERATING MODEL

From application-centric to intent-centric computing

Most people frame AI as a force that replaces applications. We think the more consequential effect is that AI replaces interfaces. Today, software is organized around screens, menus, buttons, forms, and workflows because those artifacts were designed for keyboards and touchscreens — they are concessions to the limits of the input layer, not features anyone actually wants. Once AI can understand natural human input, those constraints begin to dissolve. Instead of clicking, navigating, searching, and opening applications, a user can simply express intent, and in many situations the application effectively vanishes: the user interacts directly with the outcome they wanted.

That is not a better UI bolted onto the same operating model. It is a new operating model for computing, and it is the reason we think this shift deserves an operating system layer of its own rather than a feature shipped inside an existing one. It is also the conviction that guides what we build on top of Hover OS: our interfaces, products, and content are designed for people who talk and gesture to an intelligent system, not for people clicking, typing, or tapping their way through menus.

Why the physical world changes the architecture

Voice, gesture, and presence become most powerful precisely where a person is not sitting behind a keyboard: moving through a museum, an airport, a hospital, a factory floor, a classroom. Traditional operating systems were never designed for that environment. Desktop operating systems assume one user, one device, one session, and a screen positioned in front of a fixed point. Even mobile operating systems, for all they liberated computing from the desk, still assume one phone, one identity, and one personal experience. Physical, shared environments require the opposite: many users, shared context, persistent state, location awareness, and environmental intelligence that outlasts any single person's session. The architecture has to change because the input paradigm has changed.

"A platform built for one user cannot simply be scaled up to serve many — the assumptions are load-bearing, not cosmetic."

05 · THESIS

The Hover OS thesis

We believe the next operating system opportunity is not built around a device. It is built around human presence in physical space, and around the inputs that previous operating systems were never architected to support: gesture, voice, movement, location, contextual awareness, and shared presence among multiple people at once. Rather than treating AI as a feature that sits on top of an existing operating system, Hover OS treats AI as the mechanism that converts natural human behavior into system commands for everyone in a shared environment simultaneously.

In that framing, AI is not the product. AI is the enabler. The breakthrough — and the platform opportunity — is the new input layer itself, the one that lets people converse with computing instead of commanding it, and specifically the version of it built for many people in one place rather than one person with one device.

Agnostic at both ends

Hover OS sits between the sensors that read a room and the systems that respond to it — closer to a spatial kernel than to an application. It reads the room, infers intent, and dispatches the resulting activity directly into the computing system.



ARM and x86: complementary, not competing

ARM handles efficient sensor nodes at the edge; x86 handles heavy aggregation where data streams converge. Hover OS abstracts both, so enterprise buyers never get locked into a single silicon vendor's roadmap, pricing, or supply constraints — the same independence it maintains on the sensor side.

Two qualifications, stated plainly

Version 1 of Hover OS runs on a specific edge compute module and a specific product line of depth camera. That is a consequence of building lean rather than a change in architecture: a single reference configuration lets us ship, tune a real-time pipeline against known silicon, and prove the model in the field instead of spending early capital on breadth no customer has asked for yet. Agnosticism is the design principle and the roadmap; the range of supported hardware widens in step with adoption and resources.

We also expect to build or partner on hardware products running Hover OS. A first-party device removes the integration burden that stalls early adoption of any new platform layer — the buyer gets a system that works on arrival rather than a sensor procurement project

— and it gives us the full-stack control that the latency budget already favours. Windows shipped on other companies' boxes and Apple built its own; both owned their input era. For us the device is a market-entry vehicle for the operating layer, not a retreat from it.

Why the platform layer survives AI-generated software

The strongest objection to building an operating system in 2026 is that AI now writes software on demand, so any software business has a short half-life. We think that objection is right about applications and wrong about operating layers. Applications can be sloppy and still run; they are generated, discarded, and regenerated cheaply. Infrastructure cannot be. Nobody regenerates the network stack, the encryption path, the transit layer, or the operating system on the fly, because the swarms of agents that will build everything above them need those layers to be stable, standardized, and predictable in order to navigate across industries, IT stacks, jurisdictions, and regulatory regimes.

Eventually — plausibly in an AGI or ASI era — even those foundations will be in constant, self-managed flux, with no versioning and no release schedule. That is years away. In the interim, the durable plays are the stable substrates that generated software gets written against, and a spatial input layer is one of them.

06 · THE FIELD

Where Hover OS sits relative to the field

The space is no longer empty. Incumbent headsets own the personal device. Gesture peripherals proved single-user tracking — and Ultraleap's trajectory, ending with its patents at a licensing firm and its team absorbed into ROLI's music-learning products,

shows that best-in-class tracking sold as a peripheral did not become a platform. Oblong and Archetype AI each validate the premise that intent in physical space is the next interface. None of them is the operating layer.

	Incumbents APPLE / META	Gesture peripherals ULTRALEAP	Hover OS HOVERVERSE
USERS	Single user	Single user at a desk	**Multi-user, shared room**
HARDWARE	Personal headset	Proprietary hardware sensor	**Hardware-agnostic kernel**
OUTPUT	Floating XR windows	Hand-off to a third-party app	**Direct intent execution**

07 · CANDOR

The open problems

Two problems sit between this thesis and the product, and we would rather state them than route around them.

01 Multi-user disambiguation

Capturing intent from one person in front of a sensor is largely solved. Attributing intent to each of forty people in an acoustically hostile museum gallery, classroom, or concourse — without personal wearables to uniquely identify and track them — is not. We have not

solved this yet, and we would rather say so than imply otherwise. We are confident it is solvable, and the techniques to solve it are in active development. Anthropometrics and voice differentiation sit at the core of that work. We will describe the approach publicly once the appropriate IP protections are in place. Solving attribution without wearables is, in our view, the central technical moat in this category, which is why we are protecting the work before we publish it.

02 Hardware standardization

Spatial computing hardware is still heavily customized and combinatorial, which is a fair argument against writing an operating system for it today. It is also the condition in which the desktop and mobile eras began. We treat the absence of a standard as the opportunity rather than the obstacle: the abstraction layer that normalizes heterogeneous sensors is precisely the asset, and the platform that defines that interface tends to become the standard rather than wait for one. In the interim it also shapes how we go to market: shipping our own hardware running Hover OS is the fastest way past the integration barrier that keeps buyers from adopting a new platform layer, and every deployment on our own reference stack teaches us what the abstraction ultimately has to normalize. Breadth follows adoption; it does not precede it.

03 Processing latency

A third constraint shapes the architecture rather than threatening the thesis. Inferring intent from vision and audio is easy for a model and hard to do inside a real-time budget — a room that answers in two seconds is not an interface, it is a delay. That pushes the hot path toward deterministic, efficient processing rather than general-purpose model calls, and it is the reason the moat may reasonably extend into the hardware stack. Controlling more of the pipeline buys both the efficiency and the defensibility.

THE QUESTION FOR INVESTORS

Who owns the interface when computing moves off the screen and into the room?

PUNCH CARD	KEYBOARD + MOUSE	MULTI-TOUCH	SHARED PHYSICAL SPACE
IBM	Windows	iOS / Android	Unclaimed

Not how large AI becomes as a compute category — but who translates human intent into action where many people share one physical, persistent, intelligent space. That layer is unclaimed. Hover OS is built to own it, and now is the time to back it.

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